

AMIS 1012

Ethics in Computing



Chapter 2: Computer and Internet Crime



Learning Objectives

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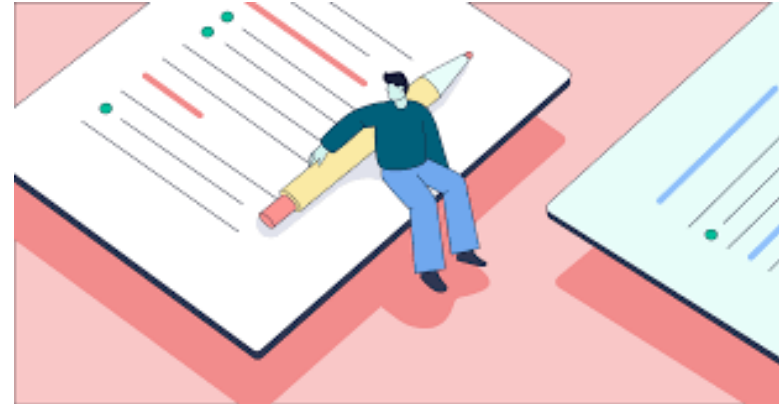
1. Examining the worms and viruses that influence Internet use and security.
2. Recognizing critical computer crime concerns.



Chapter Outline

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1. Introduction to Cybercrime
2. Computer User Expectations
3. Reliance on Commercial Software
4. Security Attacks
5. Hackers and Crackers
6. Malicious Insider
7. Cybercriminals





1.Introduction to Cybercrime

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- **Cybercrime** is a form of **illegal activity** that takes place over the **Internet**. It refers to any criminal operation carried out using modern technology.
- **Unlawful Internet operation** in which a **device** is used as a weapon, goal, or both.
- Cybercrime can be classified into two types:
 - those that cause **malicious harm**.
 - those that cause **accidental harm**.





1. Introduction to Cybercrime

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- Cybercrime often encompasses a broad variety of destructive behavior.
- Example of cybercrime:
 - **Illegal entry, illegal intrusion, and computer abuse**
 - offenses against information technology (IT) infrastructure.
 - **Data theft**
 - **Cyber bullying**
 - poses a danger to a person's physical protection, involves force, threats, or manipulation to compel someone to do something against their will.
 - shows hatred or discrimination toward those protected groups.



1. Introduction to Cybercrime

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- Example of cybercrime (cont.):
 - **Distribution of worms or viruses**
 - **Hacking** for the intent of obtaining financial or personal details
 - **Unintentional damage**
 - A dissatisfied employee creates a “harmless” virus that causes company disruption.
 - Although it does not cause the same immediate financial harm as stealing confidential or financial records, it also has a financial impact due to wasted work hours and the resources spent to correct the issue.



2. Computer User Expectations

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- Computer technology makes it **faster and less costly to capture more data** than was previously possible on paper.
- The increased utilization of electronic technology would allow for the capture of a greater variety of document.





2. Computer User Expectations

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- Transactions and information are fine-grained. It would be **possible to record and preserve all the individual modifications made** to a case file, as well as the final contents.
- Any person who “ touches” (creates, shows, copies, or modifies) a record is tracked by content/ document management systems. It’s possible that the audit trail metadata would be very large.



2. Computer User Expectations

- The **efficient resolution of computer issues is crucial** for users to quickly return to work. This puts significant pressure on tech assistance desks to respond promptly to user requests. However, this urgency sometimes leads to delays in verifying users' identities or confirming permissions for requested actions.
- Furthermore, users often share their login credentials with colleagues who have lost theirs. This practice can encourage unauthorized access to sensitive data and information systems by individuals who lack proper permissions.



2. Computer User Expectations

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- Factors contributing to a spike in computer-related vulnerability incidents:
 - Vulnerability rises as **complexity** rises.
 - The computational world has grown in complexity. Hundreds of millions of lines of code link computers, networks, operating systems (OSs), software, Web pages, switches, routers, and gateways.
 - As more devices are connected to a network, the number of potential entry points grows, raising the risk of security breaches. New risks are introduced by expanding and changing systems.



2. Computer User Expectations

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- Factors contributing to a spike in computer-related vulnerability incidents (cont.):
 - Keeping up with the **speed** of technical change, performing a successful ongoing evaluation of new security challenges, and implementing ways to deal with them is extremely challenging.



3. Reliance on Commercial Software

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- An **exploit** is a computer attack on a system that **takes advantage of a flaw in the system**. Poor system design or implementations are frequently the cause of this attack.
- When a security flaw is found, software engineers work rapidly to produce and distribute a “fix,” or **patch**, to remedy the problem. If a patch is not installed, the user risks a security breach.



4. Security Attacks

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- 2 types of security attacks:
 1. **Passive attacks**
 2. **Active attacks**



4. Security Attacks

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1. Passive attacks

- Attempts to acquire or utilize information from the system **without harming the system's resources**. A successful attack tries to **change the system's resources or disrupt its activity**.
- The objective is to **intercept the information being exchanged**.
- When coping with passive threats, the focus is on **avoidance** rather than identification.

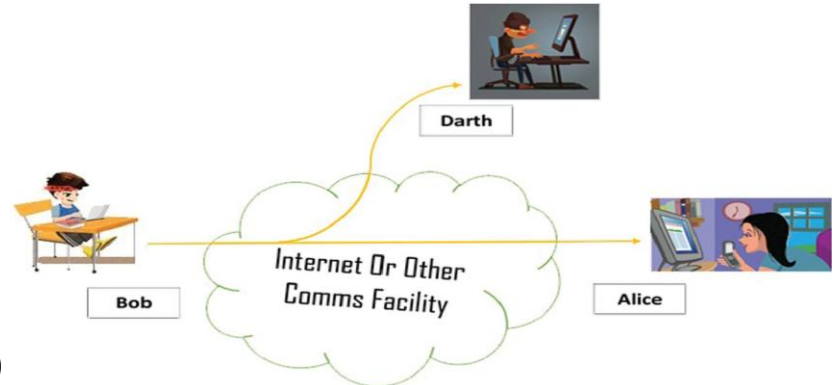
4. Security Attacks



- Example of passive attacks:

- **Eavesdropping**

- A phone conversation, an electronic mail address, or a transmitted file may all include relevant or private information.
 - Attacker may secretly listens to a conversation or communication that is not intended for the attacker.



4. Security Attacks

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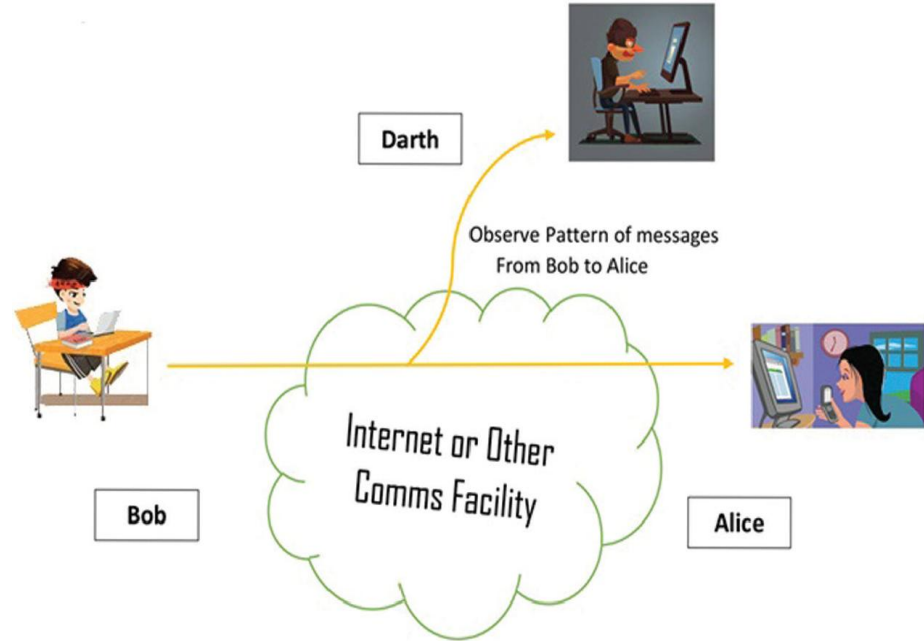


- Example of passive attacks (cont.):
 - **Traffic monitoring**
 - Attacker may **monitor the location and identity of transmitting hosts, frequency and duration of messages sent** and received.
 - Difficult to detect, as it does not change the data.
 - The sender or the recipient may not aware that a third person has read the messages or detected the traffic pattern.
 - Cryptography - possible to avoid these attempts from succeeding.

4. Security Attacks



- Example of passive attacks (cont.):
 - **Traffic monitoring**



4. Security Attacks

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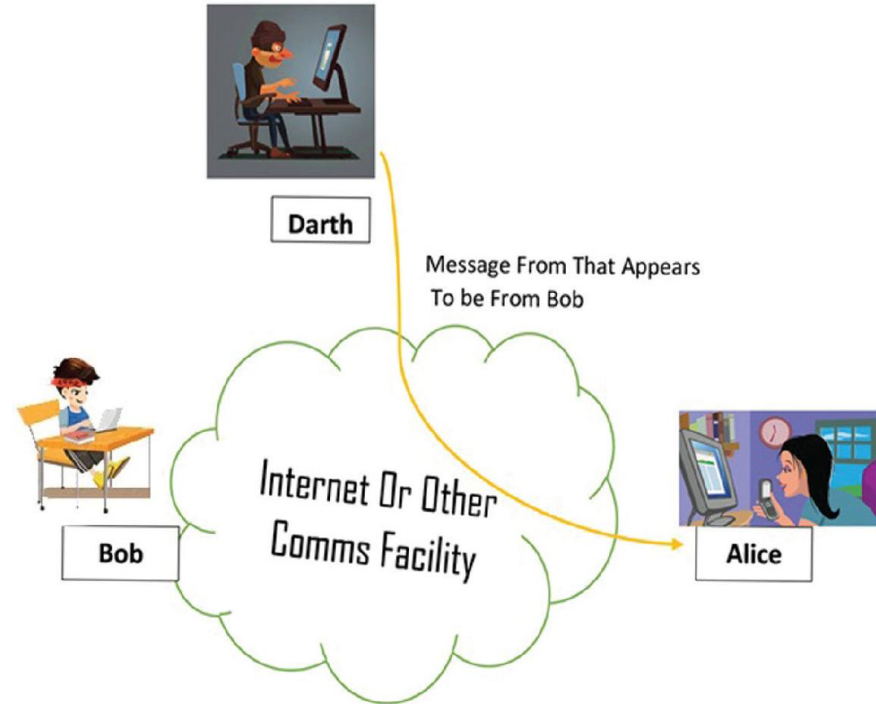
2. Active attacks

- Involves **tampering with the data stream or creating a fake stream.**
- 4 groups of active attacks (cont.):
 - **Masquerade** - One person pretends to be another.
 - For example, a trusted entity with restricted permissions to acquire more privileges by impersonating a trusted entity with those privileges.

4. Security Attacks



- 4 groups of active attacks (cont.):
 - **Masquerade**



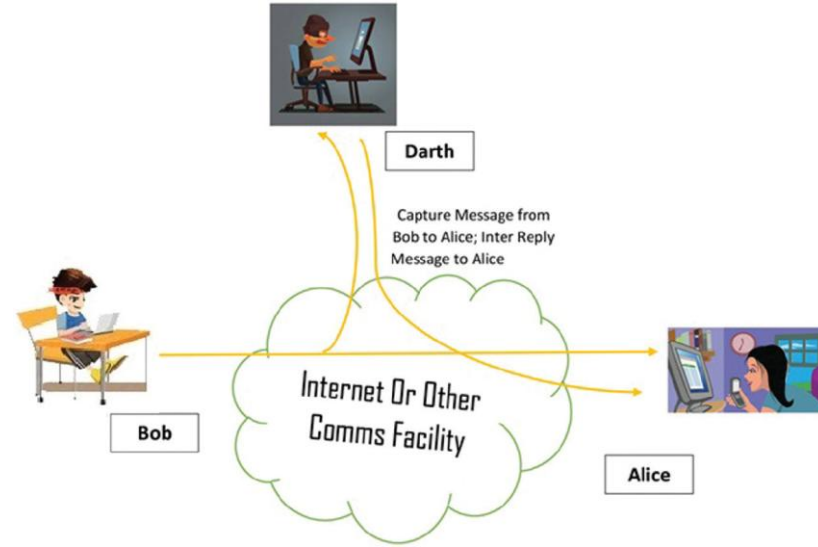
4. Security Attacks



- 4 groups of active attacks:

- Replay**

- Passive recording of a data device and eventual retransmission to achieve an unauthorized result.



4. Security Attacks

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- 4 groups of active attacks:
 - **Message alteration**
 - Some portion of a legitimate message is altered, or messages are delayed or reordered.
 - **Denial-of-service**
 - Attackers flood the system with fake requests or data in order to make a service, website, or server unavailable



4. Security Attacks – active attacks

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- **Virus**
 - A virus is a piece of **computer code** that enables a system to **behave in an unexpected and usually unpleasant manner**.
 - Most viruses carry a “payload,” or malicious content, which causes the device to behave abnormally.
 - When a computer user opens an infected email attachment, an infected program is installed on the machine or visits infected websites, and the infection spreads to other computers.



4. Security Attacks – active attacks

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- **Virus**

- Viruses propagate through the network, a disc, file sharing, or compromised email attachments as the device or document to which they are connected is passed from one device to another.
- Viruses propagate as a result of the **actions** of the “infected” machine user.
- Viruses may have a wide variety of consequences, from slightly irritating to **destroying data or applications** to inducing **denial- of-service (DoS) situations**.



4. Security Attacks – active attacks

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- **Virus**
 - Example: Macro virus
 - Attackers corrupt documents with a program written in an application macro language (such as Visual Basic or VBScript).
 - A macro virus infects a user's application and then inserts itself in all subsequent documents created by the application. The virus may delete all data on the receiver's hard disks if the email recipient clicks on the file.



4. Security Attacks – active attacks

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- **Worms**
 - A worm is a harmful software that **duplicates itself in a computer's active memory.**
 - Worms differ from viruses in that they can **transmit without the aid of people**, sometimes by emailing copies of themselves to other machines.
 - Worms are isolated programs that do not need a host program or human assistance to spread. Worms take advantage of a flaw on the target device or use social engineering to persuade users to run them.



4. Security Attacks – active attacks

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- **Trojan Horse**

- A Trojan disguises itself as a **genuine application** or file. It attempts to persuade you to install malware on your computer by tricking you into doing so. A Trojan will perform the purpose for which it was designed such as damage, disrupt, steal, or otherwise harm your data or network.
- Trojans, unlike computer viruses and worms, **cannot reproduce themselves.**



4. Security Attacks – active attacks

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- **Botnet**
 - **Collection of computers** linked to the Internet that has been **infected with malware** and may be controlled by hackers.
 - Cybercriminals get access to these computers by infecting them with specific Trojan viruses, and then installing command and control software to carry out large-scale destructive activities.



4. Security Attacks – active attacks

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- **DDoS (Distributed Denial of Service) Attack**
 - DDoS attack is a concerted **attempt to disrupt the normal operations of a targeted server or network** by **overwhelming** the object or its neighboring networks with Internet traffic.
 - DDoS attacks operate when they use a huge number of compromised machines as attack traffic sources. These networks are made up of **malware-infected servers and other appliances (such as IoT devices) that can be manipulated remotely by an attacker.**



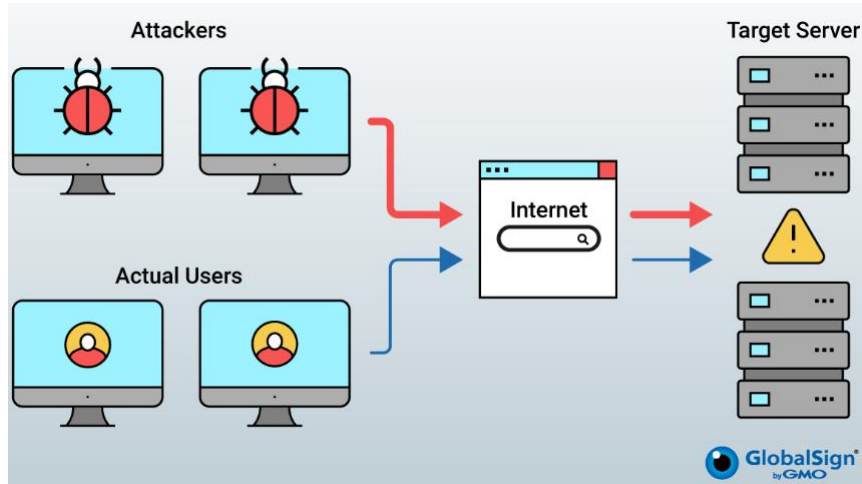
4. Security Attacks – active attacks

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- **DDoS (Distributed Denial of Service) Attack**
 - Individual robots are known as bots (or zombies), and a botnet is a collection of bots. The attacker may guide the assault by delivering remote instructions to each bot if a botnet has been created.
 - When a botnet attacks a victim's server or network, **each bot sends requests to the target's IP address, causing the server or network to become overloaded and resulting in a DoS attack.** It may be **difficult to differentiate** attack traffic from regular traffic since each bot is a genuine Internet device.

4. Security Attacks – active attacks

- **DDoS (Distributed Denial of Service) Attack**
 - DDOS attacks are carried out with networks of Internet-connected machines.



Source:

<https://www.globalsign.com/en/blog/ae/what-ddos-attack-and-how-prevent-it> (2022)



4. Security Attacks – active attacks

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- **Rootkit**
 - Rootkit is a collection of software tools that **allows a threat actor to remotely access and manipulate a device** or other machine.
 - Attackers may utilize the rootkit to execute programs, access logs, monitor user activity, and change the computer's settings.





4. Security Attacks – active attacks

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- **Spam**
 - Spam is **unrequested**. It's usually promotional, **sent to a huge number of people**, and it comes whether or not you requested it.





4. Security Attacks – active attacks

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- Type of Spam:
 - **Spam in the form of an email**: standard spam. It **logs up your mailbox** and diverts your attention away from the emails you want to post.
 - **Spam on social media**: spammers can take advantage of the growing social nature of the Internet, distributing their **spam through fake profiles on common social networking sites.**



4. Security Attacks – active attacks

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- Type of Spam (cont.):
 - **Spam on mobile phones**: it is a spam in the shape of an SMS message. Some **spammers use push alerts** in addition to spam text messages to attract your attention to their deals.
 - **Spam text messages**: It is similar to email spam, except it's a lot faster. Spammers **use instant messaging apps** such as WhatsApp and Telegram to spread their messages.



4. Security Attacks – active attacks

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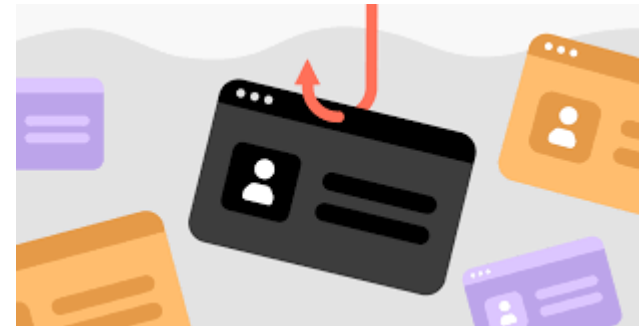
- Type of Spam (cont.):
 - Search engine optimization (SEO) spam, also known as spam indexing, is the **use of SEO techniques to boost the spammer's website's search results.**



4. Security Attacks – active attacks

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- **Phishing**
 - Phishing is a social engineering attack that involves obtaining sensitive information from individuals, such as login passwords and credit card details.
 - Social Engineering - A hacker acts as a trustworthy person and convinces a victim to establish an account, send an instant message, or send a text message.





4. Security Attacks – active attacks

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- **Techniques used in Phishing**
 - Phishing scams through email
 - Make phishing communications that appear just like **emails from a fake business**. Because they utilize the same wording, fonts, labels, and signatures, the communications seem genuine.
 - Attackers will try to persuade consumers to react by creating **a feeling of urgency**.
 - Inside-message connections mimic their legal counterparts but usually have a **misspelled domain name or additional subdomains**.



4. Security Attacks – active attacks

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- **Techniques used in Phishing**
 - Phishing with a specific purpose
 - **Targets a single individual or business.** It is a more advanced form of phishing that necessitates in-depth awareness of an organization's power system.

4. Security Attacks

– active attacks

Life cycle of phishing kit



1.

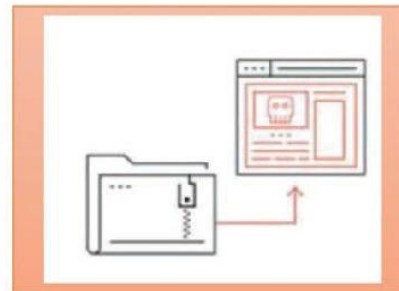
The Legitimate Website
is Cloned

2.

The Login Page is
Changed to Point to a
Credential-Stealing Script

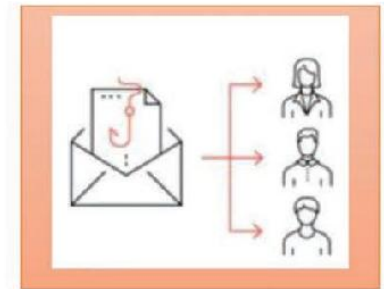
3.

The Modified Files are
Bundled into a Zip file to
Make a Phishing Kit



4.

The Phishing Kit is Uploaded
to The Hacked Website, Files
are Unzipped



5.

Emails are sent With
Links Pointing to The
New Spoofed Website



5. Hackers and Crackers

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- **Ethical Hackers**

- Ethical Hackers **explore the limits of computer networks** and see how they can obtain entry and how far they can go. They **have a clear knowledge of IT and security features**, and their inspiration derives in part from a need to learn more.
- Valid licenses, such as CEH certificates, are carried by ethical hackers. Ethical hackers use programs or hacks to test a network's security and vulnerabilities. They work with a company to help secure their data and possess Internet security skills.



5. Hackers and Crackers

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- **Crackers**
 - Cracker, is also know as malicious hacker, is someone who **maliciously hacks into or otherwise compromises the device functionality** of remote computers. Crackers kill vital records, deny lawful user service, or create issues with their targets after gaining unauthorized access.
 - If they discover a loophole, they simply erase or harm the records. Crackers are corrupt and want to profit from illegal activities.

6. Malicious Insider

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- Current or former employee, contractor, or other business partner who **intentionally exceeded or misused authorized access to an organization's network, system, or data** in a way that jeopardized the data's confidentiality, integrity, or availability.
- Frauds often require involvement, or coordination, between an individual and a third party.



7. Cybercriminals

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- Cybercriminals are **motivated by financial gain to break into business servers** to rob, often by converting funds from one account to another, leaving a hopelessly difficult path for law enforcement officers to trace.
- Cybercriminals snatch and resell credit card numbers, personal names, and mobile phone IDs, among other types of cyber theft. They can afford to pay huge amounts of money to buy the technological skills and access they need from dishonest insiders because the opportunity for monetary gain is too great.

7. Cybercriminals

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- Hacktivists
 - Hacktivism is the use of malware to accomplish a political or social objective.
 - To advance such political or social agendas and to threaten or compel a country, a cyber attacker uses computer-based assaults on other devices or networks.



7. Cybercriminals

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- Cyber terrorists
 - Cyber terrorists have more radical aims.
 - Cyber terrorists **employ tactics to kill or interrupt networks in order to inflict damage rather than collect intelligence.**
 - Example: advanced cyber terrorists use machines to hack into dams or air traffic control networks, wreaking havoc and jeopardizing not only millions of lives but also national security. However, for all of the doomsday scenarios, there has yet to be a single case of true cyber terrorism.



Question?

"Ethics is knowing the difference between what you have a right to do and what is right to do." - Potter Stewart



Disclaimer:

The slide content has been sourced from the following reference books and Internet resources. Refer to the following list for more details and further reading:

1. Awari, G.K., & Warjekar, S.V. (2022). *Ethics in Information Technology: A Practical Guide*. CRC Press, Taylor & Francis Group.
2. PyNetLabs (2024). *Difference between Active Attack and Passive Attack*. <https://www.pynetlabs.com/active-attack-and-passive-attack/>

Please note that the slides' content is subject to changes and updates.